

# Ganesh Gupta

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## EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Provisioned and operated multi-cloud **GPU clusters** (AWS, GCP, CoreWeave) on **NVIDIA CUDA** hardware with **InfiniBand/RoCE** networking for long-running training and high-availability inference.
- Wrote systems-level **C++** and **Go** for **scheduling, resource allocation, and fault-tolerant placement** under GPU-scarcity constraints.
- Built **observability** and **autoscaling** for **vLLM** inference via **Prometheus/Grafana**; ran **distributed training** through **Slurm**-based HPC scheduling, keeping fleet utilization up through node loss.
- Built **multi-agent** system via **OpenAI Agents SDK + MCP** tool-use handling **millions of daily transactions** at **99%+ uptime**.
- Drove **agentic software development** with **Claude Code** and **Codex** across the team, accelerating refactors, test generation, and code review.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Extended **vLLM**-class inference framework internals with **kernel-level development (CUDA/Triton)**, **distributed KV cache**, and **speculative decoding** to raise LLM serving performance.
- Cut p99 inference latency **420ms to 140ms** across **50K+ daily requests** by moving to async **vLLM** and **TensorRT-LLM** serving on **GCP**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Profiled compute and memory bottlenecks** across **Python/C++** inference services and optimized batching, tensor movement, scheduling, and concurrency; raised throughput **2.4x** and cut **GPU memory 27%**.
- Designed **Python/C++** model-serving infrastructure with async request handling, batching, and automated failure recovery; reached **99.9% availability** and cut p99 latency **32%**.
- Built and operated **25+** production services in **C++** and SQL-backed **distributed systems** for **1M+ users** at **99.9% uptime** on **AWS**.
- Built an automated **model-evaluation and deployment-gating** framework covering quality, latency, and regression checks; cut validation time from **6 hours to 90 minutes**.

## PROJECTS

Distributed KV Store / VPC Control Plane Simulator

[github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion VPC control plane simulator modeling subnet routing and security group enforcement.

RootCause AI

[github.com/ganeshhgupta/rootcause-ai](https://github.com/ganeshhgupta/rootcause-ai)

- LangGraph multi-agent** system for automated **incident triage** and **root cause analysis**; routes alerts to specialized sub-agents and synthesizes actionable resolution briefs.

Watch-your-LLM

[github.com/ganeshhgupta/watch-your-llm](https://github.com/ganeshhgupta/watch-your-llm)

- Production LLM eval platform with rubric-based scoring, **drift detection**, and latency tracking; **FastAPI** backend, **React/TypeScript** dashboard, **PostgreSQL**.

## EDUCATION

University of Texas at Arlington | *MS Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *BTech Computer Science*

Jul 2016 – Jul 2020

## SKILLS

**Languages:** C++, Python, Go, SQL, Java

**GPU & Performance:** CUDA, Triton, TensorRT-LLM, vLLM, Parallel and Distributed Computing, CPU/GPU Profiling, Speculative Decoding, Quantization

**AI & Agents:** PyTorch, Multi-GPU Training, OpenAI Agents SDK, MCP, LangGraph, Claude Code, Codex

**Systems:** Linux, Slurm, InfiniBand/RoCE, Kubernetes, Docker, Prometheus/Grafana, AWS, GCP, Data Structures and Algorithms

## CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer